

Cloud Computing Technologies

DAT515 - Fall 2026

Course Information

Prof. Hein Meling



WHERE
THE **HECK**
IS MY DATA?

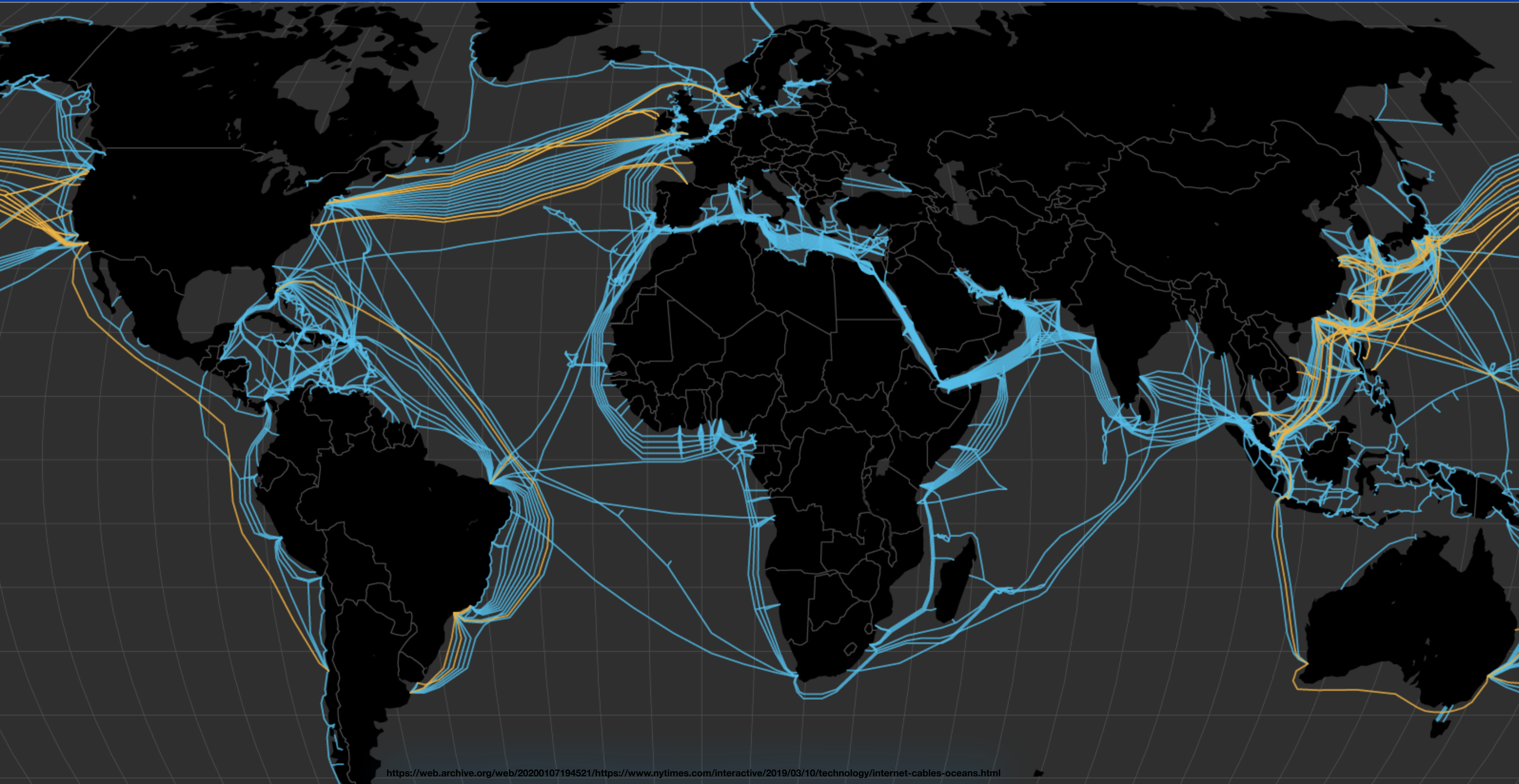


IT IS
UP THERE IN
IN THE CLOUD.



‘People think their data is
in the cloud.’

It's not!



About us

Course Instructor: Prof. Hein Meling

- hein.meling@uis.no
- Office: KE D427 (no office hours)
 - Talk to me between lectures
 - Come by the office — open door ...
 - Send email to book an appointment
- Research interests: Distributed Systems
 - Fault tolerance, availability, security, reliable storage, blockchain systems, etc...



Teaching Assistants

- Jayachander Surbiryala
 - Office KE D424
 - jayachander.surbiryala@uis.no
- Uliana Sechkar
- Viacheslav Nikolaichuk





Reliable Systems Lab



Who are you?

- What's your study program?
 - Computer Science
 - Data Science
 - Others?
- Who is working as a developer?
 - What do you do?
- Experience using cloud platforms:
 - AWS, Azure, Google Compute, Others?
- Experience using cloud technologies:
 - Docker
 - APIs for online services
 - GitHub API, Discord API,...
 - Kubernetes?

Logistics

Recommended prerequisites

- Git and GitHub proficiency expected
- Operating Systems
- Databases
- Computer Networking
- Web Programming
- Advanced Programming

GitHub

- Course information
- Lecture plan
- Lab plan
- Project information
- Reading material (links)

dat515.github.io/

github.com/dat515-2026/assignments

QuickFeed

- Submit assignments to GitHub
- Some assignments have tests
- View results of assignments
- Lab approval

<https://uis.itest.run/dat515>

*Sign up using your
GitHub account*

Enroll in DAT515

*Consider creating a separate
GitHub account for course work*

Discord

Signup Links

- Announcements
- Online help
- Online lab approval (only for special cases)
 - Contact Jayachander on Discord to organize online approval

Week 34: 2026

	Mon 17/8 ↔	Tue 18/8 ↔	Wed 19/8 ↔	Thu 20/8 ↔	Fri 21/8 ↔
08:00					
09:00	Forelesning 08:15-10:00 Room: KE E-101 Staff: H. Meling			Forelesning 08:15-10:00 Room: KE E-101 Staff: H. Meling	Laboratorium 08:15-10:00 Room: KE E-456 Staff: J. Surbiryala
10:00					
11:00				Forelesning 10:15-12:00 Room: KE E-456 Staff: J. Surbiryala	
12:00					
13:00					
14:00					
15:00					
16:00					

Week 34: 2026

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12:00					
13:00					
14:00					
15:00		Forelesning 14:15-16:00 Room: KE E-166 Staff: H. Meling			
16:00					

Weekly Schedule: Week 37 - 45

Week 37: 2026

	Mon 7/9 ↔	Tue 8/9 ↔	Wed 9/9 ↔	Thu 10/9 ↔	Fri 11/9 ↔
08:00					Laboratorium 08:15-10:00 Room: KE E-456 Staff: J. Surbiryala
09:00					
10:00					
11:00				Laboratorium 10:15-12:00 Room: KE E-456 Staff: J. Surbiryala	
12:00					
13:00					
14:00					
15:00					
16:00					
17:00			Laboratorium 16:15-18:00 Room: KE E-456 Staff: J. Surbiryala		
18:00					

Reduced TA Capacity

- Prefer to get approvals down before Week 37
- Priority: assistance over approvals

Week	Date	Day	Lecture Topic	Lecturer	Slides
34	Aug 17	Monday	Introduction to Cloud Computing	Hein	course-info cloud-intro
34	Aug 20	Thursday	Cloud APIs and Microservices Concepts	Hein	cloud-apis
34	Aug 20	Thursday	Introduction to Go Programming (lab)	Hein	go-intro
35	Aug 24	Monday	DevOps: Best Practices	Hein	devops
35	Aug 25	Tuesday	Virtualization and Containers	Hein	virtualization
35	Aug 27	Thursday	Docker	Hein	docker
36	Aug 31	Monday	Serverless Computing	Hein	serverless
36	Sep 1	Tuesday	Guest Lecture: Terraform	Vinay Setty	[terraform]
36	Sep 3	Thursday	Kubernetes: Container Orchestration	Hein	kubernetes
36	Sep 3	Thursday	Talos Lab Intro (lab)	Jayachander	[talos-intro]
41	Oct 5	Monday	Guest Lecture: UC Berkeley	Shubham Mishra	[pirateship]

“I love deadlines. I like the whooshing sound
they make as they fly by.”

— Douglas Adams, *The Salmon of Doubt*

Lab	Topic	Grading	Approval	Submission	Effort	Deadline
1	Unix Basics and Command Line Tools	Pass/fail	Automatic	Individually	3-5 hours	August 23
2	Introduction to Go Programming	Pass/fail	Automatic	Individually	4-8 hours	August 30
3	Network Programming with REST and gRPC	Pass/fail	TA Approval	Individually	3-5 hours	September 6
4	Getting Started with Docker: Deploying a Basic Web App	Pass/fail	TA Approval	Individually	5-8 hours	September 13
5	Talos Kubernetes Cluster Setup	Pass/fail	TA Approval	Individually	6-10 hours	September 27
6	Design Document for Course Project	Pass/fail	TA Approval	Group	2-4 hours	October 4
7	Course Project	A-F	Teachers	Group	30-40 hours	November 8
8	Project Presentation Video	Pass/fail	Automatic	Group	2-4 hours	November 11

Feedback System on QuickFeed

Course Project

<https://github.com/dat515-2026/assignments/tree/main/6design>

- Start planning the project this week (read the project description)
- Prepare design doc
 - Overview
 - Architecture
 - Technologies
 - Deployment

<https://github.com/dat515-2026/assignments/tree/main/7project>

- Will be created by QuickFeed
- When you create a group
- Will be named like this:
 - <https://github.com/dat515-2026/group-name>

<https://github.com/dat515-2026/assignments/tree/main/7project>

- Project Overview
- Prerequisites
- Build Instructions
- Deployment Instructions
- Testing Instructions
- Usage Examples

- Architecture Overview
- Presentation Video
- Troubleshooting

Keep it short and to the point!!

Carefully reviewed

<https://github.com/dat515-2026/assignments/tree/main/7project>

- contract.md
 - design.md
 - report.md
 - contributions.md
 - hours.csv
- Edit these files during the project execution

<https://github.com/dat515-2026/assignments/tree/main/8youtube>

- YouTube Presentation
- Q&A Session

Course Policies and Working With AI

Group for Project

dat515.github.io/#/policy

- Start socializing!
 - Put down your phone and close the laptop during breaks!
 - Optionally: Use the #group-maker channel on Discord to connect
- Groups of two (max three)

Group for Project

dat515.github.io/#/policy

- Agree on a name
- Sign up for a group on QuickFeed
- Sign the Group Contract (6design)

Partner Problems

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- Please **tell us** and **your partner** early
 - decide to drop out of the course
 - want to dissolve the group and work alone
- To avoid problems, we suggest to organize regular
 - joint work sessions



Tell us!

Generative Models

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- Use of generative models for code generation is **allowed**.
- Must be able to **explain** the code as if you had **written it yourself**.
 - May require some discipline — like rewriting the code manually
- The code must obviously **solve the assignment** and pass a sufficient number of tests on QuickFeed.



Contributions

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- Group members must **contribute equally** to the code
- Group members are expected to be able to explain the code written by other group members.



Deadlines and Slip days

dat515.github.io/#/policy

- If you cannot make a deadline, you can use up to **10 slip days** throughout the semester, including weekends and holidays.
- It is the **date on your lab's submission on GitHub** that counts towards the slip days.
- Approval **can be done after** the deadline, but at most two labs at a time
 - and depending on TA capacity

Deadlines and Slip days

dat515.github.io/#/policy

- Slip days **cannot be used to extend** the project and presentation deadlines.
- Project and presentation deadlines are hard.

Old Working Styles

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- Pair programming
 - Alternate between coding vs watching
 - Make sure to switch user that writes and commits code to GitHub
- Code Review
 - Each group member creates a branch
 - Create Pull Request
 - Partner reviews your code before it is merged into the main branch

New Working Styles

dat515.github.io/#/policy

- The PIR model: Cost-saving strategy
 - **P**lan: use a *powerful* model; carefully review and revise the plan
 - **I**mplement the plan: use a *cheap* model
 - **R**eview and fix code: use a *powerful* model
- Manually review the code (GitHub PRs), not just the artifact

When to Commit

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- Commits should represent a **logical unit of work**.
 - A bug fix, a new feature, a refactoring, etc.
 - It is up to you to decide what constitutes a unit of work, but it should not be too large.
- *Unacceptable to commit an entire lab assignment in one commit.*

Commit Messages

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- Describe the changes you have made.
- Should follow the Scoped Commits format.

```
<scope>: <description>
```

```
[optional body]
```

```
[optional trailer(s)]
```

- `<scope>` — the subsystem, area, or module that the commit touches
- `<description>` — a short description of the changes made
- `[optional body]` — detailed information about the changes
- `[optional trailer(s)]` — additional metadata about the commit

Commit Messages

dat515.github.io/#/policy

- Describe the changes you have made.
- Should follow the Scoped Commits format.

```
i2c: virtio: mark device ready before registering the adapter
virtio_i2c_probe() synchronously probes child i2c drivers on the bus,
but peripherals may use the bus at probe for tasks like reading a chip
id. The vhost-user-i2c backend stalls at such probes unless DRIVER_OK
is already set before the virtqueue is first kicked.
```

Set DRIVER_OK explicitly before i2c_add_adapter(), as done for the same reason in commit [f5866db](#) ("virtio_console: enable VQs early") and commit [71e4b8b](#) ("virtio_rpmsg: set DRIVER_OK before using device").

Signed-off-by: Alexis Bouzigues <BouziguesAlexis@JohnDeere.com>

Signed-off-by: Wolfram Sang <wsa+renesas@sang-engineering.com>

Stacked Pull Requests

- Makes it easier to review large changes by organizing them into separate pull requests
- `gh stack` command

dat515.github.io/#/policy

- ☐  **teacher: add the Course Logs page** ✓
 #1572 opened 54 minutes ago by meling Member · 6/6
- ☐  **web: add the GetCourseLog RPC** ✓
 #1571 opened 1 hour ago by meling Member · 5/6
- ☐  **all: apply go fix across the repository** ✓
 #1570 opened 1 hour ago by meling Member · 4/6
- ☐  **logging: add the course-scoped log sink** ✓
 #1569 opened 1 hour ago by meling Member · 3/6
- ☐  **errors: phrase internal error strings as the operation attempted** ✓
 #1567 opened last week by meling Member · 2/6
- ☐  **logging: migrate backend to structured, context-aware slog** ✓
 #1566 opened last week by meling Member · 1/6

Study Resources

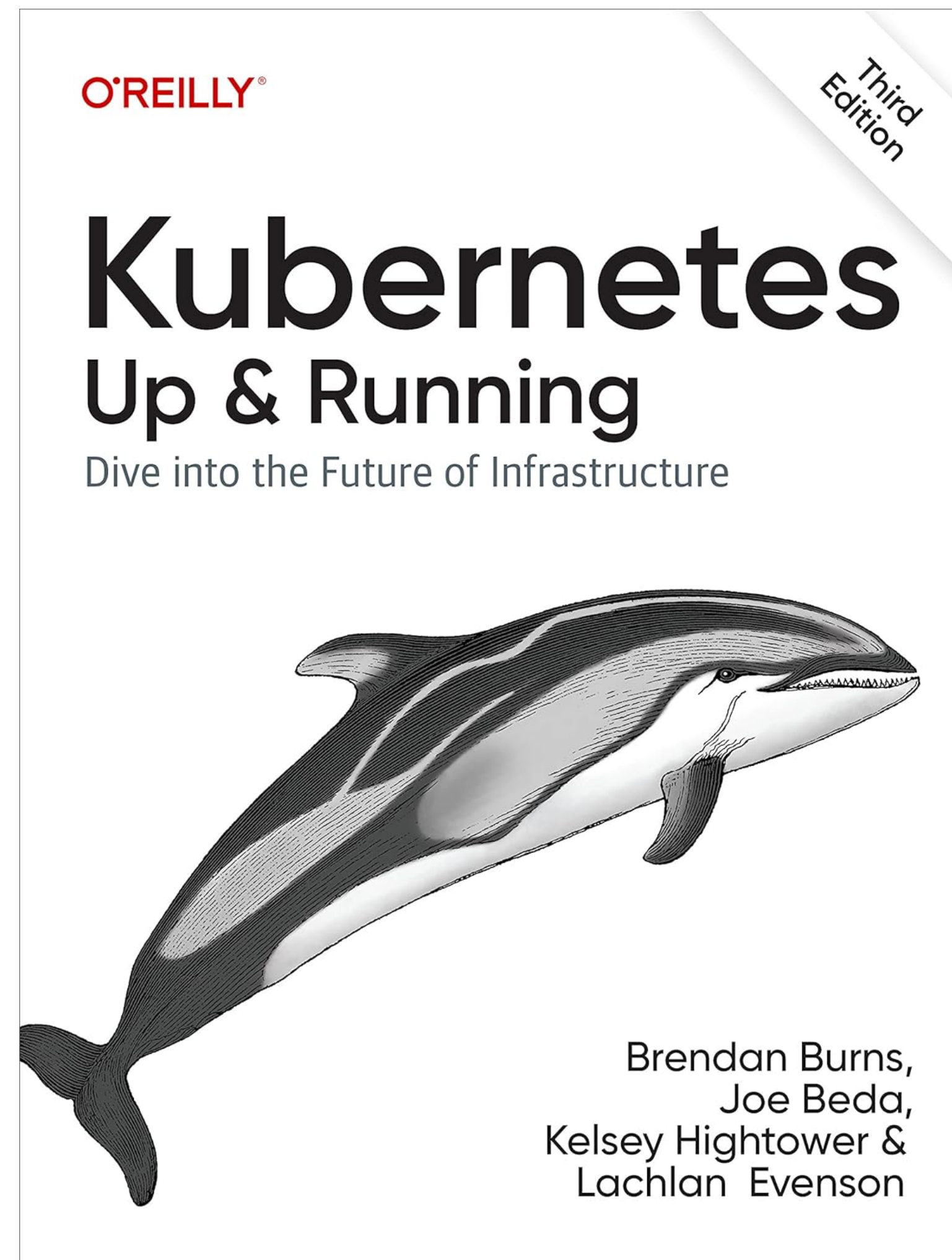
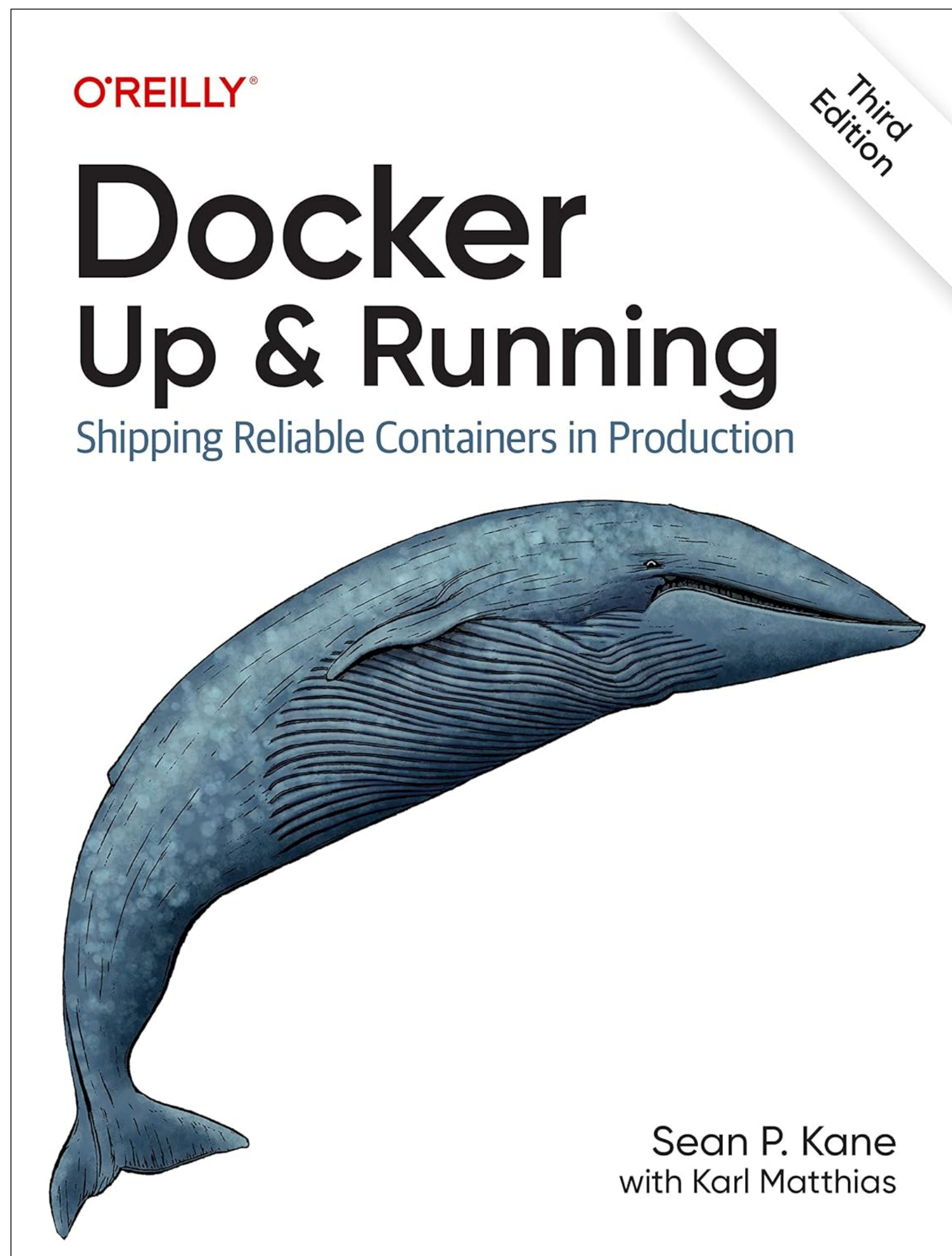
<https://dat515.github.io/#/reading-material>

<https://dat515.github.io/#/papers>

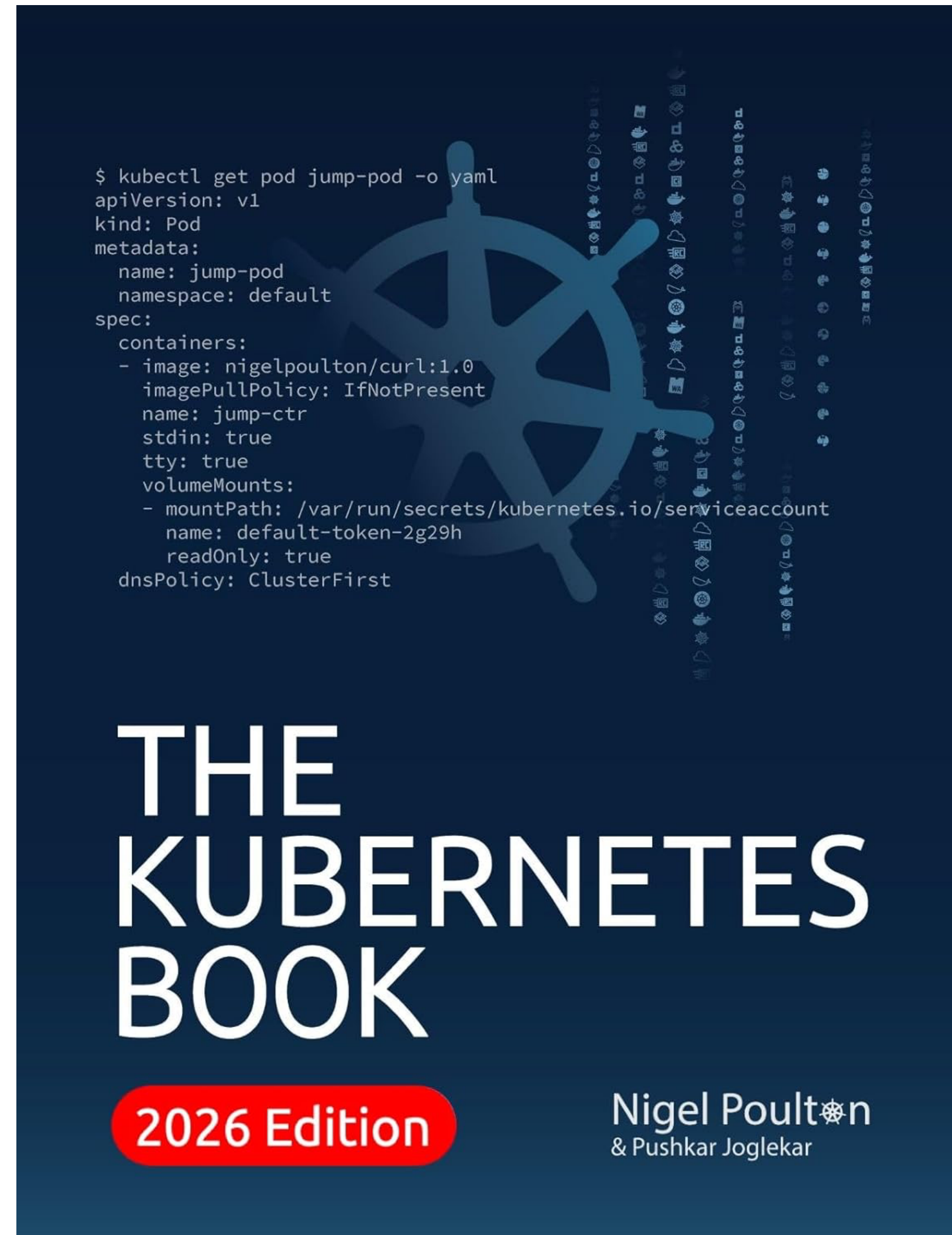
- Some overview and research papers
- Tutorials and Blogs
- Lab assignments
- Project Information
- Lecture slides

Too Many Models and Tools

- Claude, ChatGPT, Gemini, Grok, Kimi, GLM, Qwen, or GitHub Copilot
 - Command line interfaces
 - Separate Apps
- Editors: VSCode, Cursor, Zed, Antigravity, and many others
- Multiplexers: OpenCode, Supacode, Superset, Cmux, ...
- Terminals: Ghostty



- Not mandatory reading, but they are nice if you like reading books.
- Online resources are sufficient to cover the course's topics.
- Links are included in slides and assignments.



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- <https://docs.docker.com/>
- <https://kubernetes.io/docs/home/>

Links

- Official web page: go.dev
- [A Tour of Go](#)
- [Effective Go](#)
- Other [Learning Resources](#)
- [Standard library](#) and [Style guide](#)

The Go Programming Language

Alan A. A. Donovan
Brian W. Kernighan



We want your feedback
on the course

Questions?